Fireworks Algorithm for the Multi-satellite Control Resource Scheduling Problem

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***Abstract*—In this study, fireworks algorithm (FWA) for the multi-satellite control resource scheduling problem (MSCRSP) is presented. FWA is a meta-heuristic method and widely used in continuous problems while MSCRSP is a constrained and large scale combinatorial problem. The key points of FWA are to define a suitable neighborhood structure for launching the local search procedure and to find a metric for quantifying the disparity between solutions. Three kinds of neighborhood structures are presented and the best fitted one is picked. Due to the speciality of this problem, each solution is transformed into a binary vector, and Hamming distance is adopted for defining disparity metric. The experimental results demonstrate the proposed FWA is more competitive than those commonly used methods.**

I. Introduction

The multi-satellite control resource scheduling problem (MSCRSP) is a kind of large combinatorial optimization problem which involves assigning a number of resources (satellites, ground stations and time windows) to missions during a given time period. The main objective of the scheduling is to maximize the scheduled task requests or minimize the unsatisfied ones under a set of constraints.

According to the height of satellite orbit, there are three kinds of satellites, including the high-earth-orbit (HEO) satellites, the middle-earth-orbit (MEO) satellites and the low-earth-orbit (LEO) satellites. Since the control standard and resource requests of the HEO and MEO satellites are quite different from the LEO satellites, and the scheduling of the HEO and MEO satellites are relatively simple, these satellites are not selected as the researching subjects and we only focus on the LEO satellites in this paper.

A lot of works have already been dedicated to the similar problems. Traditional optimization techniques are unsuitable for MSCRSP since it is a NP-complete problem. Many researchers have turned their attentions to the intelligent optimization field. A constraint-based iterative-repair method (IR) was proposed for the GERRY scheduling and rescheduling system [1], and this method had been tested at the Kennedy Space Center (KSC). IR begins with a complete but possibly flawed solution and iteratively modifies or repairs the assignments to improve the overall schedule. Several algorithms on the satellite range scheduling problem are compared [2], and the genetic algorithm yielded the best performance on larger difficult problems. A genetic method, called Genitor, was presented and it performed well on a broad range of problem instances studied in the paper [3]. It uses the rank of fitness to maintain a population of solutions and replaces the worst one with the single child solution generated at one iteration. A tabu search heuristic was described for the problem of selecting the requests to be satisfied under operational constraints and the quality of the solutions were evaluated using an upper bounding procedure based on column generation [4]. Another tabu search heuristic was presented through studying the similarity between graph coloring techniques and satellite range scheduling problems [5]. Numerical experiments showed the heuristic was very competitive, robust, and outperformed those algorithms based on permutation. In [6], the problem was formulated as a multiprocessor task scheduling problem and a Lagrangian version of the relax-based MIP heuristic was also developed. This heuristic was shown to be equivalent to a sequence of maximum weighted independent set problems on interval graphs. A guidance-solution based ant colony optimization was proposed to avoid premature convergence for MSCRSP with the aim of minimizing the working span [7]. Two heuristic algorithms, namely, ISDR and IDI, were proposed by analyzing the dynamic properties of satellite scheduling [8]. An invariant ant colony optimization with two-stage pheromone updating was presented and it was applied to solving this problem [9].

Fireworks algorithm (FWA) is a swarm intelligence algorithm for producing good-quality solutions of continuous optimization problems [10]. This algorithm can process linear, non-linear and multi-model test functions and is suitable to implement in parallel. Most important of all, FWA has a good convergence property and can always find the global optimal solutions. These virtues make it be recognized as one of the best intelligent optimization algorithms. The application of FWA includes: non-negative matrix factorization [11], image identification [12], Spam Detection [13] etc.

However, FWA is seldom used in combinatorial optimization problems. The difficulties can boil down to two reasons: 1) the neighborhood is difficult to define. Unlike the continuous

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problem, there is no clear concept of neighborhood structure in combinatorial problems. This makes the explosion sparks and mutation sparks hard to generate. 2) the metric to measure the disparity between solutions is also hard to define. Although the neighborhood structure is constructed on the previous stage, two different permutations may result in the same scheduled tasks. So, the disparity should be redefined and quantified.

The rest of this paper is organized as follows. In section 2, the optimization model is established and the basic FWA is introduced. Then, the proposed algorithm for MSCRSP is presented in section 3. In section 4, the computational investigations are demonstrated to illustrate the effectiveness of proposed method. The conclusions of this paper are summarized in the last section.

II. Preliminaries

Control system, composed of some ground stations and a command center, plays a vital role in space system. The responsibility of the system is to track, monitor, and control the flight orbit, flying position as well as the function of subsystems in spacecrafts. The LEO satellites need to contact the ground stations several times during an assigned time period to exchange information or accept instructions. Some satellites can be served by the stations located in different places, while some can only be served on a specific one. These communication can be established through the time windows generated when the satellites fly over the ground stations.

It is becoming more challenging each year to generate schedules manually because of the large and growing number of satellites and budgetary pressure that limits the construction of new stations. In order to serve the tasks as many as possible, a feasible and relatively good schedule method is urgently needed.

A. Optimization model

1) Scheduling object: If the tasks and resources are s-elected as the scheduling components in MSCRSP, a large number of decision variables emerge. These variables along with numerous constraints of the problem will complex the established model. Unlike other papers, [7] firstly introduced the visible arc as the scheduling object. Data provided by Xi'an Satellite Control Center include the satellite number, station number, start time and end time of the time window as well as the arc type in each visible arc. The information contained in a visible arc is complete and unique, and it is also selected as the scheduling object in this paper. A visible arc can be described by a five-tuple, $a_i = \{sa_i, st_i, ts_i, te_i, or_i\}$, where a_i is a visible arc, sa_i is the satellite containing this visible arc, st_i is the service station, ts_i is the start time of a_i, te_i is the end time of a_i, and or_i is arc type, when a_i is in the raising orbit, $or_i = 1$, otherwise, $or_i = -1$. Arc set $A = \{a_1, a_2, \ldots, a_n\}$, where n means the number of visible arcs in the scheduling horizon. The satellite set $S = \{s_1, s_2, \ldots, s_{|S|}\}$, and $|S|$ means the number of satellites involved with the tasks. The station set $M = \{m_1, m_2, \ldots, m_{|M|}\}$, where $|M|$ means the number of stations used in the scheduling horizon and each station may have several equipments with the same functionality. The task set $J = \{j_1, j_2, \ldots, j_{|J|}\}$, where $|J|$ is the number of tasks.

2) Constraint analysis: For each task request, it should be accomplished between its release time and due time, arc type should be satisfied as well. On top of this, the equipments in this station should support the service of that kind of satellite, otherwise, even there are visible arcs, the communication could not be established. The visible arcs should also satisfy some intrinsic constraints:

- Equipment unit capacity constraint. An equipment can only serve one single satellite until the task is finished. In other words, once an arc is selected, those arcs that their time windows overlap the selected one and are expected to be served on the same equipment should be abandoned. Supposing two visible arcs a_i and a_j are processed on the same equipment, if $st_i = st_j$ and $[ts_i, te_i] \cap [ts_j, te_j] \neq \emptyset$, equipment unit capacity constraint is violated.
- Satellite unit capacity constraint. A satellite can only be served by one single equipment at a time, which means if the satellite has the possibility to communicate with several different ground stations at one position, it can exchange data with only one of them. For two visible arcs a_i and a_j, if $sa_i = sa_j$ and $[ts_i, te_i] \cap [ts_j, te_j] \neq \emptyset$, satellite unit capacity constraint is violated.

3) Model construction: The time needed to serve a task is usually considered as the length of the corresponding visible arc for the LEO satellite and a task j is characterized with a release time r_j, a due time d_j, its corresponding satellite sa_k as well as ground station st_l, which means the task should be processed in the time slot $[r_j, d_j]$ with the appointed satellite and station. As a result, the available arc set for task j is defined as follows:

$$Availble\_arc(j) = \bigcup_{i \in C} a_i$$

where $C = \{i | sa_i = sa_k, st_i = st_l, [ts_i, te_i] \subset [r_j, d_j]\}$, and only one of them can serve for task j.

The solutions are encoded as the permutations of the visible arcs. The scheduling procedure can be depicted as follows: for a given permutation $x = \{x^1, x^2, \ldots, x^n\}$, x^i is picked from the left to the right to serve the tasks sequentially. If a_i conflicts with those selected arcs or its corresponding task has been satisfied, then a_i is discarded. Otherwise, it will be chosen to serve the task. The objective of the scheduling is to serve the tasks as many as possible under the constraints aforementioned. The objective function is given as follows:

$$f(x) = n - |discarded\_arc|$$

where $discarded\_arc$ is a set composed by those discarded arcs and $|discarded\_arc|$ is the number of arcs contained in this set. $f(x)$ is the fitness of solution x, and it stands for the scheduled task number.

B. Fireworks algorithm

Fireworks algorithm was first proposed in [14] for global optimization of complex functions in 2010. Then, a lot of modified algorithms are proposed, which can be found in [15] [16]. The basic FWA is an iterative methodology, where each iteration consists of three stages: setting off N fireworks at N different locations, obtaining the locations of sparks and evaluating the quality of these locations so as to choose N new ones for the next iteration. The specific strategies of FWA can be described as follows.

1) Initialization: N solutions are selected randomly from the search space as fireworks are let off in the different positions overhead.

2) Local search: Once a solution is constructed, search scope and the number of searches are determined before local search procedures carry on. After the fireworks are exploded, a certain number of sparks are generated in a circle with the center of the fireworks. In FWA, there are two kinds of sparks been generated: explosion sparks and mutation sparks. Explosion sparks are responsible for the neighborhood search and mutation sparks are mainly intended to increase the diversity of the population.

For a minimization problem, the explosion radius and the number of explosion sparks of each firework are calculated as follows:

$$A_i = \hat{A} \times \frac{f(x_i) - y_{min} + \varepsilon}{\sum_{j=1}^{N} (f(x_j) - y_{min}) + \varepsilon}$$

$$N_i = \hat{N} \times \frac{y_{max} - f(x_i) + \varepsilon}{\sum_{j=1}^{N} (y_{max} - f(x_j)) + \varepsilon}$$

where
A_i is the explosion radius,
$\hat{A}$ is the maximum radius,
N_i is the number of sparks generated by the i^{th} firework,
$\hat{N}$ is the total number of explosion sparks,
$f(x_i)$ is the fitness of x_i,
$y_{max} = \max(f(x_i))$, and $y_{min} = \min(f(x_i))$,
ε is an extremely low number and is used to prevent the denominator from becoming zero.

After the sparks number and explosion radius are determined, for a d dimensional solution, the sparks are generated in a mechanism described in **Algorithm** 1. $U(a, b)$ stands for a number sampled from $[a, b]$ uniformly, and $deviation$ is the offset on each dimension.

In order to maintain the diversity of the solutions and enhance the local search ability, Gaussian sparks strategy is employed to produce sparks with Gaussian distribution. Here, mutation sparks refer to Gaussian sparks. The sparks number is given in advance, denoted as num. The procedure to generate mutation sparks are presented in **Algorithm** 2. $N(\mu, \sigma^2)$ represents a random number sampled from a Gaussian distribution with mean μ and variance σ^2.

3) Selection: One of the highlights of FWA is its selection strategy. Unlike those algorithms with fitness proportionate reproduction, FWA allocates its reproductive trials with the disparity between one solution and the others, and the disparity usually refers to the Euclidean distance in the continuous space. The best solution is selected to the next generation due to the holding the fittest individual strategy. Supposing the candidate set is K, roulette wheel method is adopted to choose the rest $n - 1$ solutions. The probability of selecting a candidate solution x_i is given as follows:

Algorithm 1 Generate explosion sparks for each firework

for $i = 1$ to N **do**
  for $j = 1$ to N_i **do**
    $\hat{x}_i := x_i$;
    Choose $z(z < d)$ dimensions from solution x_i randomly to construct a set DS;
    $deviation := A_i \times U(-1, 1)$;
    for each k in DS **do**
      $\hat{x}_{ik} := \hat{x}_{ik} + deviation$;
    end for
    Return $\hat{x}_i$;
  end for
end for

Algorithm 2 Generate mutation sparks for each firework

for $i = 1$ to num **do**
  Randomly choose a solution x_j from N fireworks;
  $\hat{x}_j := x_j$
  Choose $z(z < d)$ dimensions from x_j randomly to construct a set DS;
  for each k in DS **do**
    $\hat{x}_{jk} := \hat{x}_{jk} \times N(1, 1)$;
  end for
  Return $\hat{x}_j$;
end for

$$p(x_i) = \frac{R(x_i)}{\sum_{x_j \in K} R(x_j)}$$

and $R(x_i) = \sum_{x_j \in K} ||x_i - x_j||$.

The individuals with larger distance from others have a higher probability to be chosen regardless of its fitness. For a certain solution surrounded by a lot individuals, even it has a higher fitness, it gets a less probability to be selected. This strategy contributes a lot to the excellent performance of FWA.

III. FWA for MSCRSP

In this section, the FWA is extended to deal with MSCRSP in detail. As for the local search procedures, three kinds of neighborhood structures are defined and the explosion sparks and mutation sparks are generated in them. On the selection stage, the metric to measure the disparity between solutions are defined. The pseudo-code of the algorithm is given in **Algorithm** 3.

A. Local search

Since the solutions sampled from the solution space randomly are not guaranteed to be locally optimal, a better value can always be found after a local search procedure. As for the combinatorial optimization problems, [17] defines the neighborhood between candidate solutions as follows:

Definition 1: (Neighborhood). A neighborhood is a mapping $N : \varphi \to 2^{\varphi}$, that associates to each solution a set of candidate solutions, called neighbors.

$N(x)$ is a set composed by the neighbors of a solution x . Usually, the neighborhood is not defined explicitly unless an operator is chosen. The neighborhood of a solution is defined to be the set of candidate solutions that can be reached by applying the operator a given number of times. Supposing Π is the set of all permutations of the number $\{1, 2, \ldots, n\}$. π is one of the permutations, $\pi \in \Pi$. It can be written as follows:

$$\pi = \{\pi_1, \pi_2, \ldots, \pi_n\}$$

π_i is the i^{th} element in π, and $\pi_i \neq \pi_j$ $(i \neq j)$

- Swap operator

The swap operator is used to swap two adjacent elements in a permutation. If the swap operator is imposed on the i^{th} element in π, we will get π', which can be denoted as follows:

$$\pi' = \{\pi_1, \ldots, \pi_{i-1}, \pi_{i+1}, \pi_i, \pi_{i+2}, \ldots, \pi_n\}$$

- Insert operator

The insert operator selects two elements π_i and π_j randomly, not necessary the adjacent ones, and put the first element behind the second one. If $i > j$, π' obtains by imposing this operator on π, it can be written as

$$\pi' = \{\pi_1, \ldots, \pi_j, \pi_i, \pi_{j+1}, \ldots, \pi_{i-1}, \pi_{i+1}, \ldots, \pi_n\}$$

- Interchange operator

The interchange operator chooses two elements π_i and π_j randomly, and their positions are exchanged. If $i > j$, we will get π' as follows:

$$\pi' = \{\pi_1, \ldots, \pi_{j-1}, \pi_i, \pi_{j+1}, \ldots, \pi_{i-1}, \pi_j, \pi_{i+1}, \ldots, \pi_n\}$$

The explosion and mutation sparks can be generated with the same or different operators. Since MSCRSP is the maximization problem, the explosion radius and number of sparks are modified as follows:

$$A_i = \hat{A} \times \frac{y_{max} - f(x_i) + \varepsilon}{\sum_{j=1}^{N} (y_{max} - f(x_j)) + \varepsilon} \tag{1}$$

$$N_i = \hat{N} \times \frac{f(x_i) - y_{min} + \varepsilon}{\sum_{j=1}^{N} (f(x_j) - y_{min}) + \varepsilon} \tag{2}$$

Where

A_i is the explosion radius which refers to the maximum times that an operator performs on x_i;

$\hat{A}$ is the maximum times that an operator performs on a solution.

N_i is the number of sparks generated by i^{th} firework;

$\hat{N}$ is the total number of explosion sparks.

B. Selection

FWA allocates its reproductive trials with the Euclidean distances between one solution and the others, but in combinatorial problem, this kind of metric has not been defined explicitly. In MSCRSP, the different permutation may result in the same served tasks, so the disparity should be calculated based on the scheduled tasks instead of permutations.

A decision variable w is defined to indicate whether an arc a is chosen to serve a task. It can be described as a binary state variable:

$$w_i = \begin{cases} 1 & \text{if } a_i \text{ is chosen} \\ 0 & \text{otherwise} \end{cases}$$

the solution x_i is a permutation of the arcs, and it can be translated into $W_i = [w_1, w_2, \ldots, w_i, \ldots, w_n]$. As a result, the disparity between two solutions x_1 and x_2 can be defined as the Hamming distance $D(x_1, x_2)$. It can be written as follows:

$$D(x_1, x_2) = \sum_{i=1}^{n} W_1 \oplus W_2$$

and $\oplus$ is exclusive all operation.

For example, supposing there are five visible arcs in a schedule horizon, $W_1 = [0, 1, 1, 0, 1]$ and $W_2 = [0, 0, 1, 1, 1]$ are the scheduled task indexes corresponding to two solutions x_1 and x_2. $W_1 \oplus W_2 = [0, 1, 0, 1, 0]$. The disparity exists in the second and fourth elements, so the Hamming distance between this two solutions can be calculated: $sum([0, 1, 0, 1, 0]) = 2$. The distance between solution x_i and all the other solutions is denoted as $R(x_i)$.

The pseudo-code of FWA for MSCRSP are given in **Algorithm** 3.

IV. Experimental results and analysis

In this section, numerical analysis will be carried out to distinguish the best operator from others. After that, experiments are performed to illustrate the validity of FWA. The algorithms mentioned in this section were compiled in MATLAB R2012a, the experimental results were obtained on a PC with an Intel core i7-3770 3.4GHz processor and 8.0 GB of memory.

A. The problem instances

There are four instances tested in this paper. The basic information is given in table I. The visible arcs are calculated by Satellite Tool Kit (STK)[1]. The satellites and ground stations are chosen from its data base. The scheduled tasks are given by Xi'an Satellite Control Center, and scheduling horizon is a week.

[1] http://www.agi.com/products

Algorithm 3 FWA for MSCRSP

Set parameters $N, \hat{A}, \hat{N}, num$;
Construct a matrix $Matr$ to store each solution;
Generate N solutions randomly and store them in $Matr$;
while Stop criterion not satisfied **do**
  for $i = 1$ to N **do**
    A_i and N_i are calculted with formula (1) and (2);
    for $j = 1$ to N_i **do**
      Impose the selected operator A_i times on x_i and the new solution is denoted as x_i';
      $Matr := Matr \cup \{x_i'\}$;
    end for
  end for
  for $k = 1$ to num **do**
    Choose a solution x from the initial solution set randomly;
    Impose the selected operator a given number of times on x and the new solution is denoted as x';
    $Matr := Matr \cup \{x'\}$;
  end for
  Calculate the fitness of solutions in $Matr$ and pick the best one, denoted as x_{best};
  The best individual is reserved to the next iteration;
  Calculate $R(x)$ for each solution in $Matr$;
  Pick the other $n-1$ solutions with roulette wheel method based on $R(x)$;
end while
Return x_{best};

TABLE I
THE BASIC INFORMATION OF THE SCHEDULING

Problem instances	Instance1	Instance2	Instance3	Instance4
Num. of stations	3	3	3	3
Num. of equipments	6	6	6	6
Num. of satellites	16	16	24	24
Num. of tasks	532	548	801	764
Num. of arcs	913	917	1274	1281

B. The comparison of the neighborhood structures

There are three kinds of neighborhood structures mentioned in the previous section. In FWA, the same or the different operator can be used to produce sparks in local search stage. As a result, three operators can be combined up to nine groups to generate explosion and mutation sparks. In order to pick out the best group of operators, experiments are carried out. The maximum number of iterations are set to 100 and each combination is tested 20 times independently.

In Fig.1(a), (b) and (c), explosion sparks are produced by the single SWAP, INSERT or INTERCHANGE operator respectively, while mutation sparks are generated with all these three operators. the mean values of scheduled task number (STN) are presented.

From the three curves in each subgraph, conclusion can be drawn that mutation sparks generated with the INTERCHANGE operator are superior to the others. In order to find the best group of operators, the INTERCHANGE operator is employed for mutation sparks, and explosion sparks are generated with these three different operators. The comparison results are shown in Fig.2.

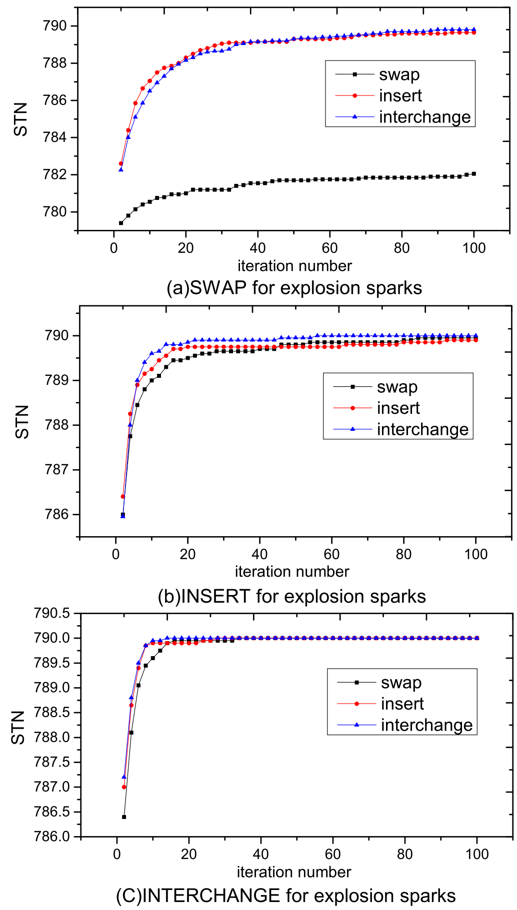

Fig. 1. Explosion sparks with the same operator

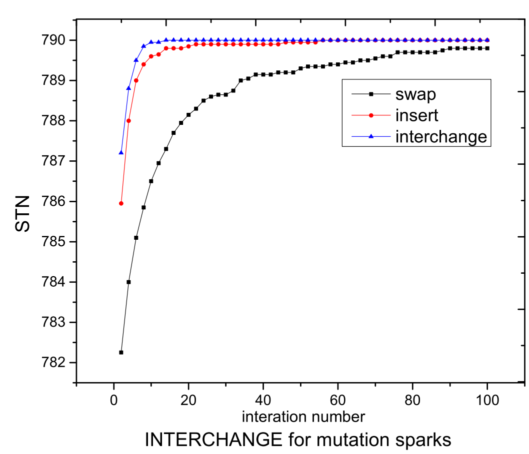

Fig. 2. Mutation sparks with the same operator

The INTERCHANGE operator still yield the best results in Fig.2. The black curve with squares in Fig.1(a) stands for the solutions sampled in the neighborhood generated with a single SWAP operator. They are inferior to those produced with INSERT and INTERCHANGE operators evidently. When a permutation is presented, the arcs are picked to serve a task from the left to the right one by one. In this way, whether an arc has the chance to be chosen to serve a task has a lot to do with its position, which means, the arc appearing in the left has a higher probability to be chosen than the one appearing in the right. The SWAP operator affects the permutation by exchanging the position of two adjacent elements, which stands for two adjacent arcs, and the probability that these two arcs are chosen is not changed significantly. The INSERT operator transforms a solution by putting an element to another position, not necessarily the adjacent ones, and this will make a difference on the selected arcs. The INTERCHANGE operator transforms a solution by altering the position of two elements, which may affect more scheduled tasks than INSERT. These characteristics explain why INTERCHANGE operator is the best while SWAP is the worst.

C. The performance of FWA

FWA is applied to four instances aforementioned. An experiment is also carried out to compare with the performance of iterative repair (IR), and genetic algorithm (GENITOR). Since the number of solutions generated at each iteration are quite different, it is meaningless to set the maximum iterations to compare the performance of the algorithms. Consequently, the runtime in each experiment is set to a constant value, and each algorithm is run 20 times for different instances independently. If the given CPU time is achieved, the algorithms stop.

Two time limits were tested: 90 seconds and 180 seconds. The best and average objective values of the solutions are presented in table II and table III, respectively. We can conclude the following points from the table. Firstly, the average values generated by FWA are the best among the three algorithms. Secondly, all the best solutions are produced by Genitor and FWA. Thirdly, for large scale problems, IR is less competitive compared with the other two algorithms.

TABLE II
COMPARISON ON FOUR INSTANCES IN 90SECONDS

	IR		Genitor		FWA	
Instance	best	mean	best	mean	best	mean
Instance1	524	521.37	**528**	526.91	**528**	**527.48**
Instance2	538	534.50	**540**	538.49	539	**538.56**
Instance3	770	759.80	**790**	787.37	**790**	**789.45**
Instance4	729	718.82	747	744.10	**754**	**750.01**

TABLE III
COMPARISON ON FOUR INSTANCES IN 180SECONDS

	IR		Genitor		FWA	
Instance	best	mean	best	mean	best	mean
Instance1	525	522.14	**528**	527.00	**528**	**527.75**
Instance2	539	534.53	**541**	539.63	**541**	**540.82**
Instance3	768	762.25	**790**	787.49	**790**	**789.73**
Instance4	730	720.69	749	745.51	**754**	**751.59**

V. CONCLUSION

In this paper, FWA is extended to solve MSCRSP. Three neighborhood structures are generated by SWAP, INSERT and INTERCHANGE operators independently. According to experimental results, the INTERCHANGE operator is the best no matter it is used to generate explosion or mutation sparks. Moreover, to make FWA work, the solutions are transformed into binary vectors, so the Hamming distance can be adopted. Compared with IR and Genitor, FWA can provide the most competitive results.

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